

Afif Haziq Bin Haris

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Professional Experience

Malaysian Communications and Multimedia Commission (MCMC), Cyberjaya, Selangor April 2026 - Present
Assistant Director, Network Security Research Department

- Engineered COSMOS, a production-grade AI-powered compliance portal serving hundreds of licensees, by building end-to-end ML pipelines for automated report analysis and scoring, reducing manual compliance review effort and enabling continuous intelligent monitoring at scale.
- Maintained a production ML-integrated content moderation system processing 300,000+ cases, by delivering new features and resolving incidents on a Django/React stack, sustaining continuous deployment cycles with zero data loss and consistent model serving reliability.
- Reduced case duplication by 95% and operationalised real-time model output monitoring by configuring Metabase dashboards to visualise content moderation statistics, surfacing insights that directly informed ministerial briefings presented in Parliament.
- Built an automated DNS compliance monitoring pipeline tracking 1,000+ blocked domains across Malaysian ISPs in real time, by implementing systematic surveillance workflows, enabling continuous data quality checks aligned with MCMC's regulatory enforcement mandate.
- Led technical validation of an LLM-powered sentiment analysis platform covering 30 million+ users, by benchmarking automated content scoring, sentiment classification, and OSINT profiling against contractual SLAs across three departments ensuring model quality and governance standards before production rollout.
- Achieved 99.99% uptime across ML-serving applications by containerising workloads with Docker and deploying a full observability stack (Grafana, Loki, Prometheus, Promtail, Wazuh) with Ansible playbooks automating the entire provisioning and environment standardisation pipeline across all environments.

Centre of Research for Cyber Security & Network (CSNET), Universiti Malaya May 2024 - October 2025
Network Security Researcher

- Increased edge inference throughput by 238% on NVIDIA Jetson Nano by engineering a lightweight CNN for Encrypted Network Traffic Classification, executing the full model lifecycle from training and XAI-based explainability (DeepSHAP) through compression (pruning, quantisation) to production deployment via PyTorch, ONNX, and TensorRT.
- Curated and published MalayaNetwork_GT on Hugging Face, a versioned benchmark dataset of 12.5 million encrypted packets across 10 classes, establishing a verified, reproducible data baseline and demonstrating data governance and lineage best practices for downstream ML workflows.
- Published and presented "A Lightweight Network Traffic Classification Using XAI" at IEEE ICOCO 2024, documenting model architecture decisions, experiment results, and reproducibility protocols, reinforcing rigorous ML documentation and reporting standards.
- Provided technical guidance to 30 undergraduate students on networking fundamentals, building cross-functional communication skills aligned with supporting data and ML engineering teams on platform usage and tooling.

Navigate Pte Ltd September 2023 - March 2024
Network Operation Centre (NOC) Intern

- Monitored and managed 1000+ global network circuits (BB, SD-WAN, DIA, VPN, 4G, IP Transit) across 150+ countries, ensuring high availability and uptime.
- Conducted operational activities within NOC, overseeing incident management, technical fault resolution, and service delivery.
- Served as the primary technical liaison for L1/L2 fault resolution, successfully mitigating 100+ technical escalations weekly via Jira while maintaining a 99% SLA adherence rate.

- Generated and presented monthly and yearly reports on NaviGate Networks, ISP services, and Facilities' status and performance.

Education

Universiti Malaya Master of Computer Science (Research Mode)	2024 - 2025
Universiti Teknologi MARA (UiTM) Arau, Perlis Bachelor of Computer Science (Hons.) in Computer Networks (CGPA: 3.5)	2020 - 2023
Universiti Teknologi MARA (UiTM) Dengkil, Selangor Foundation in Engineering (CGPA: 3.59)	2019 - 2020

Projects

- GoByte - High-Performance Network Packet Parser** February 2026
- Engineered a high-performance ML data ingestion pipeline in Go achieving 776,000 packets/second throughput at ~50 MB peak RAM, by implementing goroutine-based concurrency and streaming architecture, exporting to CSV, Parquet, and NumPy formats for direct ML/DL model integration and feature engineering workflows.
- Adversarial attacks on Continual Learning-based Intrusion Detection Systems (IDS)** January 2026
- Engineered a Continual Learning IDS using TabTransformer and Experience Replay, enabling dynamic model retraining and adaptation to zero-day threats without catastrophic forgetting validating model robustness under MITRE ATT&CK-aligned adversarial conditions (Backdoor and Label Flipping attacks).
 - Executed MITRE ATT&CK tactics (Persistence, Defense Evasion), targeting the replay buffer with Backdoor and Label Flipping attacks to quantify and harden model resilience.
- Applied Deep Learning for Intrusion Detection Systems (IDS)** December 2025
- Demonstrated the full ML lifecycle from data preparation through deployment-ready packaging using PyTorch, WandB for experiment tracking, and ONNX Runtime for model serialisation covering data cleaning workflows, reproducible experiment logging, and standardised deployment patterns for production-grade models.
- 2025 IEEE Computer Society Malaysia Chapter Webinar Series and professional Workshops** December 2025
- Conducted a session for IEEE Computer Society Malaysia on adaptive AI architectures, guiding researchers on overcoming catastrophic forgetting. Presented real-world strategies for applying Continual Learning to mitigate dynamic network security threats using Avalanche and PyTorch.

Professional Leadership

- Vice President of Postgraduate Club - Universiti Malaya
- Committee member for Postgraduate Research Symposium (PGRES) & IEEE Image Processing, Image Analysis and Real-Time Imaging Symposium (IPIARTI) - Universiti Malaya
- Speaker for Research Tools: Let's get it started - Universiti Malaya
- Facilitator of NVIDIA MONAI Image Analysis Training - Universiti Malaya
- Facilitator for Malaysia ICT Volunteer (MIV) Program by Malaysian Communications and Multimedia Commission (MCMC) - Arau, Perlis

Certificates

- CCNAv7: Introduction to Networks (Cisco NetAcad)
- CCNAv7: Switching, Routing, and Wireless Essentials (Cisco NetAcad)
- CCNAv7: Enterprise Networking, Security, and Automation (Cisco NetAcad)

- Cybersecurity Essentials by Cisco Networking Academy
- CompTIA CySA+ Courses by BlackBerry CCOE
- Cybersecurity & AI by BlackBerry CCOE

Skills

Tools: Docker, Ansible, GitHub Actions, Grafana, Prometheus, Loki, Promtail, Wazuh, Hugging Face, Linux, Git, SLURM, Nginx, Wireshark, Nmap, Jira

Frameworks: PyTorch, TensorFlow, TensorRT, ONNX, Scikit-Learn, LangChain, Pydantic AI

Data: WandB, Pandas, NumPy, Matplotlib, ChromaDB, DeepSHAP, Metabase, PostgreSQL

Programming Languages: Python, Go, Bash, SQL, Java, C++, C

Hardware: NVIDIA, Cisco, Juniper, Fortinet, Mikrotik

Language: English (Proficient), Bahasa Melayu (Native)

References

- **Ts. Dr. Muhammad Faiz Mohd Zaki** - faizzaki@um.edu.my
- **Dr. Azizi Ariffin** - mazizi@fskm.uitm.edu.my
- **Prof. Ts. Dr. Nor Badrul Anuar Juma'at** - badrul@um.edu.my